

NHS methodology compliance note

Task – 3 : Week - 4

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1. Regulatory Alignment and Professional Standards :

This analytical framework was developed in accordance with the UK Statistics Authority's Code of Practice for Statistics, emphasizing trustworthiness, quality, and value. It aligns with the NHS Data and Information Strategy (2022) by promoting the "Better, Broader, Safer" use of health data for secondary purposes.

- **RAP Principles:** The pipeline is built as a Reproducible Analytical Pipeline (RAP). This means the transition from raw data to the final dashboard is managed entirely through code, eliminating manual copy-pasting errors and providing a complete audit trail for clinical reviewers.

2. Data Protection Impact Assessment (DPIA) Summary :

While a full DPIA is a separate legal document, the pipeline incorporates the following "Privacy by Design" technical controls:

- **Anonymization through Aggregation:** The pipeline strictly adheres to the ICO Code of Practice on Anonymisation. By aggregating patient counts to the Sub-ICB level (where n is generally >5), we mitigate the risk of "jigsaw identification" (the ability to identify a person by combining multiple small data points).
- **Data Minimization & Storage Limitation (GDPR Art. 5):**
 - * **Selection Logic:** The MHSDSPipelineModule uses a "Pre-Filter" mechanism. It scans the 240,000+ raw rows in memory and only writes the specific rows required for clinical demand modeling to the permanent master_historical.csv.
 - **Retention:** Administrative columns and metadata not used in the 1,563-row master file are purged immediately upon the script's completion.
- **Data Accuracy (GDPR Art. 5(1)(d)):** The pipeline includes an automated Idempotency Check. This cross-references the Reporting_Period_Start and Geographic_Code before appending data. If a record for that specific month and

region already exists, the pipeline updates the record rather than duplicating it, ensuring the "Single Version of Truth."

3. Algorithmic Transparency and Quality Assurance :

To satisfy NHS Digital's Quality Assurance (QA) standards, the following mathematical validations are applied:

1. **Suppression Handling:** In NHS datasets, values between 1 and 5 are often suppressed with an asterisk (*) to protect privacy. Our methodology treats these as "Missing at Random" (MAR). The pipeline uses `pd.to_numeric(errors='coerce')` to ensure these non-numeric characters do not cause the forecasting models to fail, instead treating them as null values that do not artificially inflate the caseload averages.
2. **Outlier Sensitivity (Isolation Forest):** The anomaly detection module uses a Contamination Factor of 0.01 (1%). This threshold is based on the assumption that extreme prescribing variance is rare. By documenting this parameter, we ensure the methodology is "explainable"—clinical auditors can see exactly why a specific CCG was flagged for review.
3. **Temporal Validation (ARIMA):** The forecasting model utilizes Akaike Information Criterion (AIC) scoring to select the best fit. This ensures we are not "overfitting" the model to noise in the historical data, providing a more stable and realistic projection for future staff requirements.

4. Clinical Governance :

This methodology note confirms that the pipeline is a Decision Support Tool, not a decision-maker.

- **Thresholds:** Anomaly flags generated by the pipeline are intended to trigger a "Clinical Audit Request," not to automatically penalize a GP practice or Trust.
- **Feedback Loops:** Any data discrepancies identified by clinical leads during the June review cycle will be fed back into the User Correction Ledger, allowing the pipeline to improve its accuracy over time.

5. Summary of Compliance Metrics :

Compliance Pillar	Metric / Control	Verification
Privacy	Aggregation to Sub-ICB Level	Manual Audit of master_historical.csv
Accuracy	Idempotent Append Logic	image_e4b5a3.png Execution Log
Auditability	Full Python Source Code	Git / Version Control History
Reliability	< 15 Minute Reproducibility	Section 14 Reproducibility Test

6. Conclusion :

The integration of the MHSDS automation module has been executed with a "Privacy by Design" approach, ensuring that all data processing activities remain proportional, necessary, and transparent. By adhering to the UK GDPR principles of data minimization and purpose limitation, the pipeline mitigates the risks associated with secondary-use health data.

The methodology demonstrated in this audit—specifically the use of Reproducible Analytical Pipelines (RAP)—establishes a new standard for internal Trust reporting. It successfully replaces manual, opaque data manipulation with an open, code-driven process that is fully auditable by Information Governance (IG) teams.

Ultimately, this compliance framework ensures that the Trust can leverage advanced data science to improve clinical outcomes and operational efficiency without compromising patient trust or regulatory standing. The pipeline is now formally verified as compliant and ready for deployment within the May 2026 reporting cycle.